Q44 - Smart Grid Load Forecasting Optimization

Dynamic Programming Based Load Forecasting

Aim:
To analyze electricity demand forecasting using Dynamic Programming for efficient energy distribution.

Problem Statement:
An energy provider must predict electricity demand to ensure efficient distribution. Analyze how Dynamic Programming can be used for load forecasting, considering seasonal variations and peak demand.

Objectives:
• Predict future electricity demand.
• Analyze seasonal and peak load constraints.
• Improve energy distribution efficiency.
• Evaluate forecasting performance.

Dynamic Programming Approach:
Dynamic Programming stores previously computed load values and uses them to predict future demand, avoiding redundant computations.

Constraints:
• Seasonal variation in demand.
• Peak load fluctuations.
• Real-time adaptability requirements.

Algorithm:
1. Store historical load values.
2. Use previous demand patterns to forecast future loads.
3. Memoize intermediate computations.
4. Predict future demand and optimize energy supply.

Real-Time Adaptability:
The model can continuously update predictions as new load data arrives, enabling adaptive energy management.

Performance Metrics:
• Forecast Accuracy
• Mean Absolute Error (MAE)
• Execution Time

Time Complexity:
O(n)

Result:
The forecasting model successfully predicted future energy demand, helping optimize electricity distribution and reduce supply shortages.

Conclusion:
Dynamic Programming provides an efficient and scalable approach for smart grid load forecasting and energy optimization.

Output Screenshot:
(Insert terminal output screenshot here.)